# Frazier N. Baker

AI Researcher & Ph.D. Candidate

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## Professional Profile

AI researcher specializing in **agentic AI for scientific discovery**. Builds intelligent systems for complex science and engineering applications, including retrosynthesis, drug design, and end-to-end drug-discovery workflows. Experienced leading interdisciplinary research programs and deploying research software. Ph.D. Candidate at The Ohio State University with expected graduation in Summer 2027.

## Education

**The Ohio State University** — **August 2021–Present**
*Ph.D. Candidate, Computer Science and Engineering* — *Columbus, Ohio*
*Advisor: Xia Ning, Ph.D.* • *Passed candidacy July 2026*
*Part-time 2021–2023; Full-time Graduate Research Assistant since 2023*

- Dissertation in progress: *AI Agents for Retrosynthesis.*
- Coursework includes machine learning, artificial intelligence, reinforcement learning, and foundations of speech and language processing.

**University of Cincinnati** — **December 2019**
*M.S., Computer Science — Accelerated Engineering Degree BS/MS Program (ACCEND)* — *Cincinnati, Ohio*
*Advisor: Alexey Porollo, Ph.D.* • *GPA: 3.97/4.00*
*Thesis: Mining and Visualization of Amino Acid Coevolution Data*

**University of Cincinnati** — **May 2019**
*B.S., Computer Science — University Honors Student; Dean's List* — *Cincinnati, Ohio*
*GPA: 3.96/4.00*

**Ohio University Zanesville** — **August 2012–August 2014**
*High School Post-Secondary Program; GPA: 4.00/4.00* — *Zanesville, Ohio*

**Tri-Valley High School** — **May 2014**
*Honors High School Diploma; GPA: 4.00/4.00* — *Dresden, Ohio*

## Research & Professional Experience

**The Ohio State University** — **August 2023–Present**
*Graduate Research Assistant, Department of Biomedical Informatics* — *Columbus, Ohio*

- Developed **agentic AI** systems for drug discovery applications, including retrosynthesis, drug design, and end-to-end drug-discovery workflows.
- Developed and demonstrated a framework for designing, constructing, and testing **multi-agent systems** for multi-objective retrosynthesis planning.
- Designed, implemented, and tested an **LLM Agent-as-a-Judge** framework for constrained retrosynthesis planning under practical safety constraints, the first AI framework for retrosynthesis in that setting; this work received a **Best Paper Award** at AIAS 2025.
- Architected and trained a multi-agent reinforcement-learning system for synthon completion in retrosynthesis; published in *Journal of Chemical Information and Modeling.*
- Collaborated with **interdisciplinary teams** of medicinal chemists, computational chemists, biologists, computer scientists, and pharmaceutical-industry partners; mentored junior students and researchers.
- Organized journal-club-style lab meetings and lab retreats, coordinating student and faculty schedules.

**Georgia Tech Research Institute, Advanced Concepts Laboratory** — **July 2022–October 2023**
*Research Scientist II* — *Dayton, Ohio*

**Georgia Tech Research Institute, Advanced Concepts Laboratory** — **June 2019–July 2022**
*Research Scientist I* — *Dayton, Ohio*

- Architected, designed, built, and tested AI and ML solutions across diverse application domains, including vehicle autonomy,

radar, mis/dis-information, and chemical and biological threat detection.
- Fine-tuned **language models** to identify and annotate types of extremism within information-warfare contexts according to a literature-based ontology.
- Developed and maintained research applications using Python, C++, ReactJS, Docker, and GitLab CI/CD DevOps pipelines.
- Designed and implemented low- and no-code systems for studying neural-network architectures and transfer learning on radar data.
- Captured and executed more than **$1M** in externally funded research for autonomy **modeling and simulation** and an AI contest.
- Led hybrid and remote teams of up to **12 researchers**; supervised 10 undergraduate interns and mentored full-time new hires and junior researchers.
- Contributed to proposals generating more than **$6M** in external research funding and more than $75K in internal research funding.
- Initiated and led an organization-wide grassroots AI-focused virtual journal club, leveraging connections across laboratories and divisions.
- Led and participated in **Agile** development teams using issue boards, daily stand-ups, sprint meetings, and point estimation.
- Presented to high-profile government sponsors, including AFRL, NRL, DTRA, Big Safari, and AFLCMC.

**Cincinnati Children's Hospital Medical Center** **2015–2016; 2017–2019**
*Student Researcher, Center for Autoimmune Genomics and Etiology* *Cincinnati, Ohio*

- Designed and implemented a well-cited visualization platform for amino-acid coevolution data.
- Built **deep-learning models** for protein metal-binding-site prediction and analyzed datasets containing over 695K records.
- Developed **proteomics** and **toxicogenomics** tools using JavaScript, Perl, and R; first author on **three peer-reviewed journal articles**.
- Named Employee of the Month in February 2016.

**Kinetic Vision** **2016–2017**
*Software Development Co-op, Web Development Team* *Evendale, Ohio*

- Developed full-stack web applications for corporate customers using Ruby on Rails, JavaScript, SCSS, ActiveRecord migrations, MVC design, and AWS cloud deployment.
- Collaborated in Agile development teams using issue boards, daily stand-ups, biweekly sprint meetings, and time/difficulty estimation.
- Designed backend database schemas and migrations; partnered with graphic designers to implement user-friendly frontend designs.

**Ohio University Zanesville** **January–August 2014**
*Peer Tutor, Student Success Center; Laboratory Assistant, Physics Laboratory* *Zanesville, Ohio*

- Tutored undergraduate mathematics from pre-algebra through univariate calculus.
- Built a Python visualization pipeline to download, extract, and analyze nuclear-physics data for more than 3,000 atomic nuclei; presented the work at the 2014 APS Ohio Section meeting.

## Publications

### Journal Articles

J1. **F. N. Baker**, Z. Chen, D. Adu-Ampratwum, and X. Ning. "RLSynC: Offline-Online Reinforcement Learning for Synthon Completion." *Journal of Chemical Information and Modeling*, 64(17), 6723–6735, 2024.

J2. D. Concoran, N. Maltbie, S. Sudalairaj, **F. N. Baker**, J. Hirschfeld, and A. Porollo. "CoeViz 2: Protein Graphs Derived From Amino Acid Covariance." *Frontiers in Bioinformatics*, 2021, 19.

J3. **F. N. Baker** and A. Porollo. "CoeViz: A web-based integrative platform for interactive visualization of large similarity and distance matrices." *Data*, 3(1), 4, 2018.

J4. **F. N. Baker**, M. T. Cushion, and A. Porollo. "A quantitative model to estimate drug resistance in pathogens." *Journal of Fungi*, 2(4), 30, 2016.

J5. **F. N. Baker** and A. Porollo. "CoeViz: A web-based tool for coevolution analysis of protein residues." *BMC Bioinformatics*, 17, 119, 2016.

### Peer-Reviewed Conference Proceedings

P1. **F. N. Baker**, D. Adu-Ampratwum, R. Averly, B. Yu, H. Sun, and X. Ning. "LARC: Towards Human-level Constrained

Retrosynthesis Planning through an Agentic Framework." *Tianqiao and Chrissy Chen Institute Symposium on AI for Accelerated Science*, 2025. **Best Paper Award**.

P2. R. Averly*, **F. N. Baker***, and X. Ning. "LIDDIA: Language-based Intelligent Drug Discovery Agent." *EMNLP*, 2025. **Equal contribution.*

P3. Z. Chen et al., including **F. N. Baker**. "ScienceAgentBench: Toward Rigorous Assessment of Language Agents for Data-Driven Scientific Discovery." *International Conference on Learning Representations (ICLR)*, 2025.

P4. B. Yu, **F. N. Baker***, Z. Chen*, G. Herb, B. Gou, D. Adu-Ampratwum, X. Ning, and H. Sun. "Tooling or Not Tooling? The Impact of Tools on Language Agents for Chemistry Problem Solving." *Findings of NAACL*, 2025. **Equal contribution.*

P5. B. Yu, **F. N. Baker***, Z. Chen*, X. Ning, and H. Sun. "LlaSMol: Advancing Large Language Models for Chemistry with a Large-Scale, Comprehensive, High-Quality Instruction Tuning Dataset." *Conference on Language Modeling (COLM)*, 2024. **Equal contribution.*

### Preprints and Manuscripts under Review

M1. Y. Liu, R. Averly, Avery Meyer, **Frazier N. Baker**, Ruoxi Gao, Daniel Adu-Ampratwum, and Xia Ning. "MFG: Multi-turn Feedback Guided Molecular Optimization" *arXiv:2610.18851*, 2026.

M2. T. Nguyen **F. N. Baker**, R. Gao, D. Adu-ampratwum, and X. Ning. "MFG: Multi-turn Feedback Guided Molecular Optimization" *arXiv:2610.18851*, 2026. (Under review at ARR October 2026.)

M3. R. Gao, **F. N. Baker**, T. Nguyen, and X. Ning. "RIDE: Reference-Anchored Inference-Time Diffusion Editing for Scaffold Hopping" *arXiv:2609.35623*, 2026. (Under review at ICLR 2027.)

M4. B. Yu, B. Zhou, D. Adu-ampratwum, **F. N. Baker**, Z. Chen, R. Averly, Y. Liu, W. Gao, X. Ning, and H. Sun. "FREA: A Multi-Source Expert Benchmark for Reaction Feasibility Verification", 2026. (Under review at ICLR 2027.)

M5. **F. N. Baker**, T. Nguyen, R. Averly, B. Yu, D. Adu-Ampratwum, H. Sun, and X. Ning. "MMORF: A Multi-agent Framework for Designing Multi-objective Retrosynthesis Planning Systems." *arXiv:2604.05075*, 2026. (Under review at *Digital Discovery*.)

M6. A. Chen, **F. N. Baker**, R. Averly, S. Zhang, T. Chen, B. Liang, D. Adu-ampratwum, X. Ning, and Y. Xue. "ChemExplain: A Multimodal Benchmark for Explanatory Chemical Reasoning", 2026. (Under review at ARR October 2026.)

M7. R. Gao, J. Peng, Z. Chen, **F. N. Baker**, D. C. Kombo, J. L. Kane, A. A. Scholte, Y. Li, M. J. LaMarche, L. I. Iconaru, H.-P. Biemann, M. Hong, and X. Ning. "Generating Developable 3D Molecules via Pocket-Conditioned Diffusion and Property-Aware Optimization." *arXiv:2607.12349*, 2026. (Under review at *Nature Communications*.)

### Theses and Dissertations

T1. **F. N. Baker**. *AI Agents for Retrosynthesis.* Ph.D. dissertation in progress, The Ohio State University.

T2. **F. N. Baker**. *Mining and Visualization of Amino Acid Coevolution Data.* Master's thesis, University of Cincinnati, 2019.

## Inventions

I1. "Systems and Methods for Pocket-Based Drug Design." Provisional patent application submitted to the USPTO; assignee: The Ohio State University, 2025.

I2. "Systems and Methods for Molecule Design." Patent application submitted to the USPTO; assignee: The Ohio State University, 2025.

## Research Funding

**Multi-platform Autonomy Simulation Suite (MASS)** **2020–2021**
*ARCNET/AFRL/DoW R&D contract; Project Director / Co-PI, GTRI/GTARC* *$946K*

**Swarm and Search AI Hackathon (SSAI)** **2021–2022**
*Wright Brothers Institute/AFRL/DoW R&D contract; Project Director / PI* *~$150K*

**Title not approved for public release** **2021–2022**
*GTRI-ACL/DoW internal funding; Project Director / PI* *~$50K*

**Karat: A Visual Framework for Constructing Neural Network Architectures** **2020–2021**
*GTRI-HIVES/DoW internal early-career funding; Project Director / PI* *$25K*

## Presentations, Abstracts, & Posters

C1. **F. N. Baker** et al. "LARC: Towards Human-level Constrained Retrosynthesis Planning through an Agentic Framework." ACS Spring, Generative and Agentic Systems in Chemistry Session, 2026.

C2. **F. N. Baker** et al. "LARC: Towards Human-level Constrained Retrosynthesis Planning through an Agentic Framework." Ohio State CSE Graduate Student Visit Day, 2025.

C3. Z. Chen, **F. N. Baker**, et al. "Methodological Advances in Generative AI for Drug Discovery: An AI-Centric Perspective." AI Driven Drug Discovery Summit, 2024.

C4. **F. N. Baker**, Z. Chen, and X. Ning. "RLSynC: Offline-Online Reinforcement Learning for Synthon Completion." Ohio State CSE Graduate Student Visit Day, 2024.

C5. **F. N. Baker**, Z. Chen, and X. Ning. "RLSynC: Offline-Online Reinforcement Learning for Synthon Completion." Molecular Machine Learning Conference, 2023.

C6. **F. N. Baker**. "Title not approved for public release." GTRI IRAD Extravaganza, 2022.

C7. M. Farrell, **F. Baker**, R. Hutchins, V. Srikanth, and B. Quick. "Population-based Fever Surveillance for Early Warning of Infectious Disease Outbreaks." Chemical and Biological Defense Science and Technology Conference, 2022.

C8. R. Hutchins, **F. Baker**, C. Thomas, and M. Lavelle. "Edge Computing for Global Genomics-based Threat Detection, Assessment, and Decision Support." Chemical and Biological Defense Science and Technology Conference, 2022.

C9. **F. N. Baker**. "Karat: A Visual Framework for Constructing Neural Network Architectures." GTRI IRAD Extravaganza, 2021.

C10. **F. N. Baker**. "Karat: A Visual Framework for Constructing Neural Network Architectures." AI for Systems Engineering and Systems Engineering for AI Workshop, 2020.

C11. **F. N. Baker**, N. Maltbie, J. Hirschfeld, and A. Porollo. "Metalloproteome Landscape from the Amino Acid Covariance Perspective." 26th Conference on Intelligent Systems for Molecular Biology, 2018.

C12. **F. N. Baker** and A. Porollo. "A Covariance-based Model to Estimate Antimicrobial Drug Resistance." The Protein Society 30th Annual Symposium, 2016.

C13. **F. N. Baker** and A. Porollo. "CoeViz: A Web-based Tool for Coevolution Analysis of Protein Residues." 24th Conference on Intelligent Systems for Molecular Biology, 2016.

C14. **F. N. Baker** and A. Porollo. "Accurate Prediction of Metal Binding Sites in Proteins." 24th Conference on Intelligent Systems for Molecular Biology, 2016.

C15. **F. N. Baker** and A. Porollo. "Prediction of Metal Binding Sites in Proteins Using Coevolution Data." Great Lakes Bioinformatics Conference, 2016.

C16. **F. N. Baker** and A. Porollo. "Comparative Analysis of Gene Expression Profiles Perturbed by Lupus-inducing Drugs." Great Lakes Bioinformatics Conference, 2015.

C17. G. Popa and **F. Baker**. "Systematics of Nuclear Level Densities." International Workshop on Nuclear Theory, 2014, pp. 187–192.

C18. **F. Baker**, B. Hina, L. Schneider, and G. Popa. "Automated Evaluation of Experimental Data for Nuclear Level Densities." APS Ohio-Region Section Annual Spring Meeting, 2014.

## Professional Service

### Peer Review

- 2027: AAAI
- 2026: *Bioinformatics*; AAAI; ACM Multimedia (Main Conference and Dataset Track); COLM.
- 2025: COLM; NeurIPS; AAAI.
- 2024: BIBM; The Web Conference; WSDM; SIGIR.
- 2023: RecSys; ICDM; SIGIR.
- 2022: MICCAI; RecSys.

Conferences listed by date of meeting, not date of review. Journals listed in *italics*.

### Panels and Committees

- Curriculum Committee, Ohio State Department of Computer Science and Engineering, Autumn 2024–Spring 2026.
- Ph.D. Student Panel, Ohio State CSE Graduate Student Visit Day, 2024.

- AI Expert Panelist, Swarm and Search Artificial Intelligence Competition, 2022.

## Teaching

**(Guest Lecturer) Quantitative Methods Capstone Seminar on Machine Learning** **February 26, 2025**
*Guest Lecturer / Panelist, The Ohio State University* *Columbus, Ohio*

## Awards, Honors, and Nominations

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| Best Paper Award, Chen Institute Symposium on AI for Accelerated Science | 2025 |
| Travel Grant, Chen Institute Symposium on AI for Accelerated Science | 2025 |
| College of Medicine Next Generation Innovator of the Year Nominee, Ohio State ERIK | 2025 |
| Mantai/Mae Scholarship, University of Cincinnati EECS | 2018 |
| Eta Kappa Nu Inductee, University of Cincinnati Chapter | 2017 |
| Employee of the Month (February), CAGE, Cincinnati Children's Hospital Medical Center | 2016 |
| Cincinnatus University Scholar, University of Cincinnati | 2014–2019 |
| J. Donald Kincaid and Julianne Kincaid Scholarship | 2014–2018 |
| T. R. Murphy and Dorothy K. Murphy Scholarship | 2014–2018 |
| National Society of Collegiate Scholars Inductee | 2014 |

## Methods & Skills

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| **Research areas** | Agentic AI; generative AI; reinforcement learning; AI for science; natural language processing; bioinformatics; cheminformatics |
| **Programming** | Python; JavaScript; LaTeX; C++; C; R; MATLAB; Julia; Ruby on Rails; Perl; SQL; HTML; CSS; Bash |
| **Frameworks** | vLLM; Hugging Face; OpenAI API; Anthropic API; PyTorch; ReactJS |
| **Engineering** | AI agents; machine learning; deep learning; data visualization; Docker; cloud computing; GitLab CI/CD; Agile development; DevOps pipelines |

## Professional Memberships

Association for Computing Machinery (ACM) • American Chemical Society (ACS) • Ohio Academy of Science (OAS)
Institute of Electrical and Electronics Engineers (IEEE)

## Literary Publications

L1. **F. Baker**. "Paper" [poem]. *Zane Traces Literary Journal*, 2014.